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Static and Two-Stage SP Models

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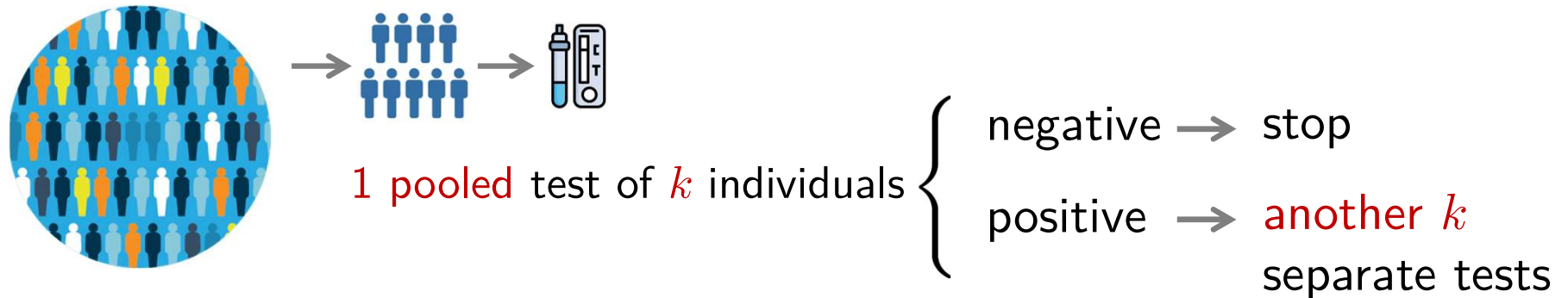
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- Examples of static SP models
- Two-stage SP models and recourse structure
- George B. Dantzig's homework
- **Exercises:**
 - Simple recourse problem formulation
 - network capacity expansion problem



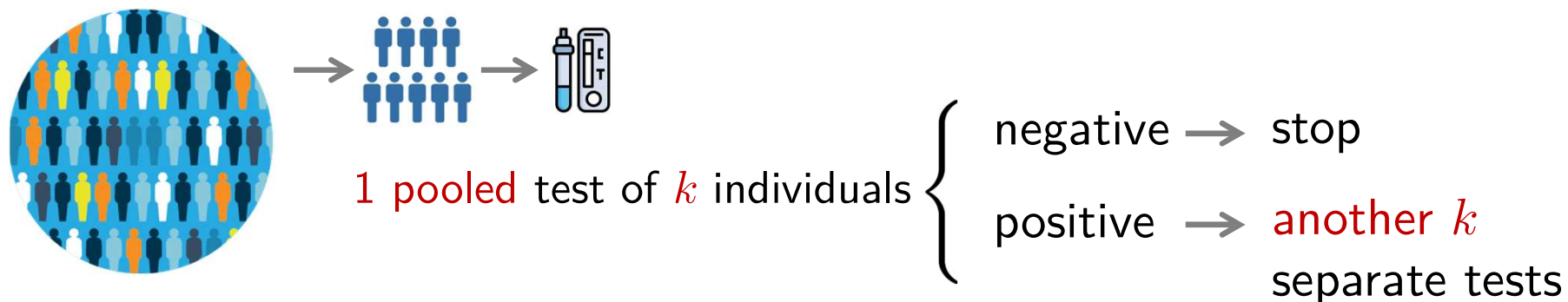
- A population of N individuals are subjected to blood testing



- Assumptions:** positive probability p and statistically independent individuals in the population
- Goal:** find the best value of k in term of the total number of tests needed to do $M(k)$
- $M(k)$ is a random variable
- The **expectation recipe:** $\min_{k \geq 1} \mathbb{E}[M(k)]$



- A population of N individuals are subjected to blood testing



$$\mathbb{E}[M(k)] \approx \frac{N}{k} \left[1 \cdot \underbrace{(1-p)^k}_{\mathbb{P} \text{ of group neg.}} + (k+1) \cdot \underbrace{(1 - (1-p)^k)}_{\mathbb{P} \text{ of group pos.}} \right]$$

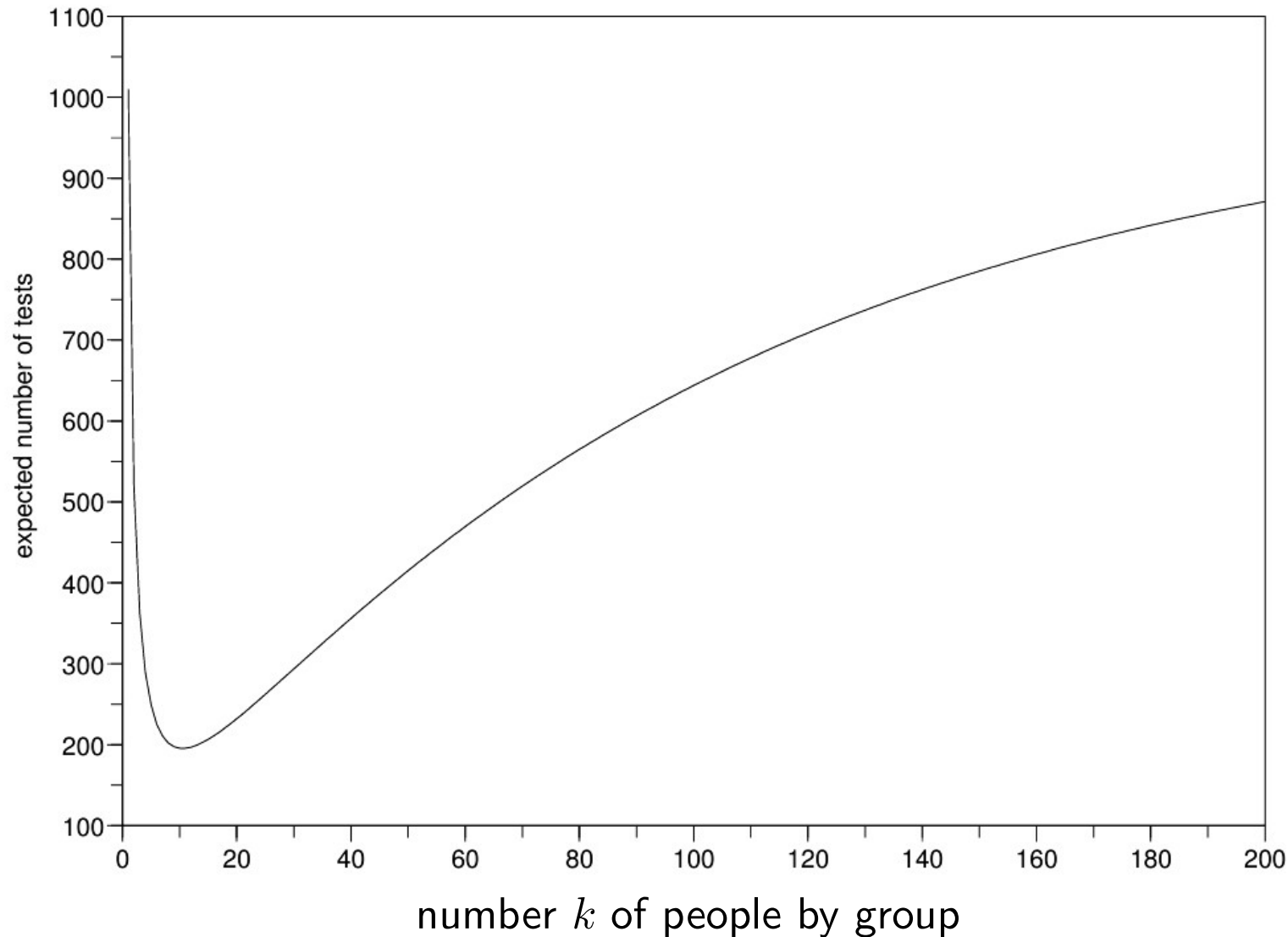


Static SO: blood testing



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Expected number of tests for $N = 1000, p = 0.01$





$$\mathbb{E}[M(k)] \approx \frac{N}{k} [1 \times (1-p)^k + (k+1)(1 - (1-p)^k)]$$

- For small p

$$\begin{aligned}\mathbb{E}[M(k)] &\approx \frac{N}{k} [1 \times (1-kp) + (k+1)(1 - (1-kp))] \\ &= \frac{N}{k} + Nkp\end{aligned}$$

- so that the optimal point is $k^* \approx \frac{1}{\sqrt{p}}$
- and the minimal expected number of tests is

$$\mathbb{E}[M(k^*)] \approx 2N\sqrt{p} < N$$

- In army practice, R. Dorfman achieved savings up to 80%, compared to making N tests

if $p = 0.01$, giving $k^* = 11 \approx 1/\sqrt{0.01}$ and $\mathbb{E}[M(k^*)] \approx 0.2N$



The static newsvendor problem



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- The “newsboy problem” is now coined the “newsvendor problem”



- The economic tradeoff: unsold copies are worthless (perishable good), but decisions are **here-and-now**
- The total cost

$$J(x, \xi) = cx - p \min\{x, \xi\}$$

is a random variable



- With the **expectation recipe**, the newsvendor solves

$$\min_{x \in \mathbb{R}_+} \mathbb{E}[J(x, \xi)]$$

- **Rationale:** to have solutions that perform the best on average with repeated implementation

$$j(x) = \mathbb{E}[J(x, \xi)] = cx - p\mathbb{E}[\min\{x, \xi\}]$$

- is a convex function, thus

$$\begin{aligned} j'(x) &= c - p\mathbb{E}\left[\frac{\partial}{\partial x} \min\{x, \xi\}\right] \\ &= c - p[\mathbb{P}(\xi \leq x) \cdot 0 + \mathbb{P}(\xi > x) \cdot 1] \\ &= c - p[1 - F(x)] \end{aligned}$$



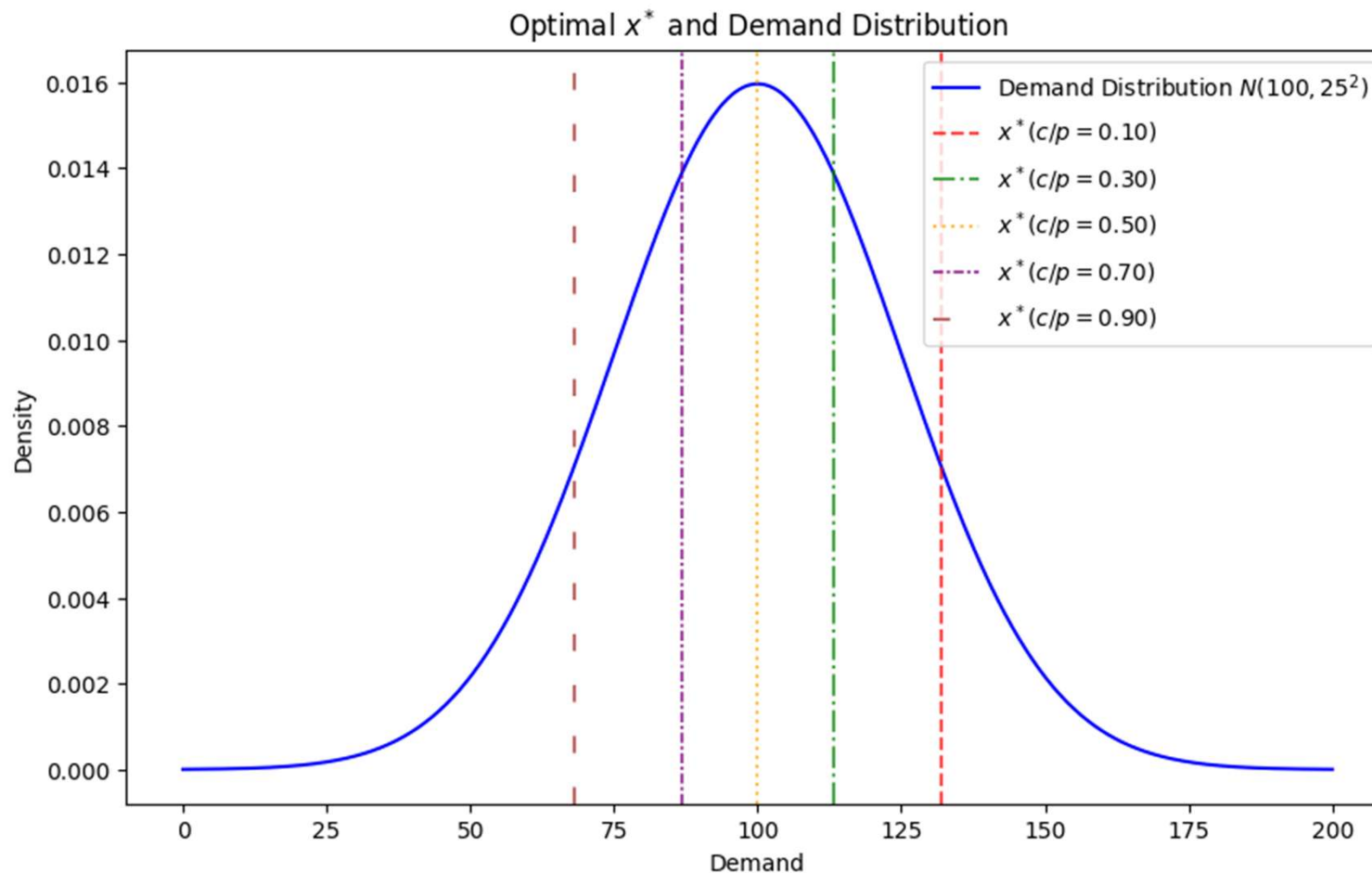
The static newsvendor problem



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- Thus

$$x^* = F^{-1} \left(1 - \frac{c}{p} \right)$$





- One-stage (static) SP model

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \mathbb{E} f_0(x, \xi) \\ \text{s.t.} \quad & f_i(x, \xi) \leq 0, \quad \forall \xi \in \Xi, i = 1, \dots, m \end{aligned}$$

- Make **one-shot decision**, then, the performance depends on the realization of uncertainty
- Often, we can make a **correction** y once the uncertainty is realized

$$x \rightsquigarrow \xi_1 \rightsquigarrow f_i(x, \xi_1) \quad \forall i$$

...

$$\rightsquigarrow \xi_k \rightsquigarrow f_i(x, \xi_k) \quad \forall i$$

$$x \rightsquigarrow \xi_1 \rightsquigarrow y_1 \rightsquigarrow f_i(x, \xi_1) \quad \forall i$$

...

$$\rightsquigarrow \xi_k \rightsquigarrow y_k \rightsquigarrow f_i(x, \xi_k) \quad \forall i$$



- Any unsold newspaper can be **returned** to the publisher with a return price $0 < r < c$ at the end of the day...

$$\min_{x \in \mathbb{R}_+} \mathbb{E}[J(x, \xi)] = \begin{cases} \min & cx - \mathbb{E}[p \cdot \text{Effective sales} + r \cdot \text{Returns}] \\ \text{s.t.} & y(\xi) \leq \xi, \forall \xi \\ & y(\xi) + w(\xi) \leq x, \forall \xi \\ & y(\xi), w(\xi) \geq 0, \forall \xi \end{cases}$$

- Write in another way:

$$\min_{x \in \mathbb{R}_+} cx + Q(x)$$

where $Q(x) = \mathbb{E}[Q(x, \xi)]$

$$Q(x, \xi) = \begin{cases} \min & -p \cdot y - r \cdot w \\ \text{s.t.} & y \leq \xi \\ & y + w \leq x \\ & y, w \geq 0 \end{cases}$$



$$\min_{x \in \mathbb{R}_+} cx + Q(x)$$

where $Q(x) = \mathbb{E}[Q(x, \xi)]$ $Q(x, \xi) = \begin{cases} \min & -p \cdot y - r \cdot w \\ \text{s.t.} & y \leq \xi \\ & y + w \leq x \\ & y, w \geq 0 \end{cases}$

- For specific ξ and $r < c < p$

$$y^*(\xi) = \min(\xi, x) \quad w^*(\xi) = \max(0, x - \xi)$$

- The 2nd stage expected value function

$$Q(x) = \mathbb{E}[-p \cdot \min(\xi, x) - r \cdot \max(0, x - \xi)]$$

- Convex, continuous, differentiable when ξ is continuous

$$Q(x) = -px + (p - r) \int_0^x F(\xi) d\xi$$



- Therefore

$$[cx + \mathcal{Q}(x)]' = c - p + (p - r)F(x)$$

- It follows

$$x^{\star} = F^{-1} \left(\frac{p - c}{p - r} \right) \geq F^{-1} \left(\frac{p - c}{p} \right)$$

- Reduce to the static solution when $r = 0$
- Should order more when $r > 0$ compared to the static case



2-stage model: general form



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- Back to our gasoline production planning example:

$$\left\{ \begin{array}{ll} \min & 2x_1 + 3x_2 + \sum_{k=1}^K p_i [7y_1(\omega_k) + 12y_2(\omega_k)] \\ \text{s.t.} & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \\ & \left. \begin{array}{l} \alpha(\omega_k)x_1 + 6x_2 + y_1(\omega_k) \geq h_1(\omega_k) \\ 3x_1 + \beta(\omega_k)x_2 + y_2(\omega_k) \geq h_2(\omega_k) \\ y_1(\omega_k), y_2(\omega_k) \geq 0 \end{array} \right\} k = 1, \dots, K \end{array} \right.$$

$$\rightsquigarrow \left\{ \begin{array}{ll} \min & c^\top x + \mathbb{E}[q^\top(\omega)y(\omega)] \\ \text{s.t.} & Ax = b, x \geq 0 \\ & T(\omega)x + W(\omega)y(\omega) = h(\omega), \text{ a.e.} \\ & y(\omega) \geq 0, \text{ a.e.} \end{array} \right.$$



2-stage model: general form



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$$\rightsquigarrow \begin{cases} \min & c^\top x + \mathbb{E}[q^\top Q(x, \xi)] \text{ Recourse function} \\ \text{s.t.} & Ax = b, x \geq 0 \\ & T(\omega)x + W(\omega)y(\omega) = h(\omega), \text{ a.e.} \\ & y(\omega) \geq 0, \text{ a.e.} \end{cases}$$

- Define $\xi(\omega) \triangleq (q(\omega), h(\omega), W(\omega), T(\omega))$ a random vector

$$Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- for fixed x and ξ , and $y \in \mathbb{R}^m$



$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases}$$

- is the **1st stage problem**, with **here-and-now** x , while for each pair (x, ξ) ,

$$Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- is the **2nd stage problem**, with recourse variable y . A simple LP for **each pair** (x, ξ)
- Recourse matrix W



- Most practical problems involve a recourse structure, allowing for **adjustments or corrective actions** after the realization of uncertainty
 - return actions in the 2-stage newsvendor model
 - buying gasoline in the market
- For one-shot static problem, recourse variables can be introduced to **handle the feasibility issue**
 - penalty formulation
 - very big penalty will force recourse variables $y \rightarrow 0$ (may have numeric stability issue...)



2-stage SLP: recourse structure



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$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases} \quad Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- What does $Q(x, \xi) = +\infty$ mean?
(infeasible recourse)
- What does $Q(x, \xi) = -\infty$ mean?
(unbounded below, rarely the case, suggesting modeling flaw, e.g., missing constraints?)
- For fixed ξ , i.e., (q, h, W, T) , when is $Q(\cdot, \xi)$ finite?



$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases} \quad Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- **Fixed Recourse:** W is **not random**
 - Simpler model, more tractability
 - The way we make recourse decisions is often fixed



- The 2-stage newsvendor model

$$\min_{x \in \mathbb{R}_+} cx + \mathbb{E}[Q(x, \xi)] \quad Q(x, \xi) = \begin{cases} \min & -p \cdot y - r \cdot w \\ \text{s.t.} & y \leq \xi \\ & y + w \leq x \\ & y, w \geq 0 \end{cases}$$

- The 2-stage production planning

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \end{aligned} \quad Q(x, \xi) = \begin{cases} \min & 7y_1 + 12y_2 \\ \text{s.t.} & y_1 \geq h_1 - (\alpha x_1 + 6x_2) \\ & y_2 \geq h_2 - (3x_1 + \beta x_2) \\ & y_1, y_2 \geq 0 \end{cases}$$



$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases} \quad Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- **Complete Fixed Recourse**

- W is not random, and
- The system $\{Wy = \chi \text{ and } y \geq 0\}$ has a solution for every $\chi \in \mathbb{R}^m$
 $\Leftrightarrow \text{pos } W := \{\chi \in \mathbb{R}^m : Wy = \chi, y \geq 0\}$ coincide with $\mathbb{R}^m$

- **Implication:** the recourse problem is **feasible for any x and $h(\omega), T(\omega)$**



$$\begin{cases} \min & c^\top x + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} & Ax = b, x \geq 0 \end{cases} \quad Q(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- **Relatively Complete Fixed Recourse**

- W is not random, and
- $\forall x \geq 0$ s.t. $Ax = b$ we have $Q(x, \xi) < \infty$ for a.e. $\xi \in \Xi$

- **Implication:** the recourse problem is feasible for **any feasible** x and almost-all possible $h(\omega), T(\omega)$



- The 2-stage newsvendor model: CFR or RCFR?

$$\min_{x \in \mathbb{R}_+} cx + \mathbb{E}[Q(x, \xi)] \quad Q(x, \xi) = \begin{cases} \min & -p \cdot y - r \cdot w \\ \text{s.t.} & y \leq \xi \\ & y + w \leq x \\ & y, w \geq 0 \end{cases}$$

- The 2-stage production planning: CFR or RCFR?

$$\begin{aligned} \min \quad & 2x_1 + 3x_2 + \mathbb{E}[Q(x, \xi)] \\ \text{s.t.} \quad & x_1 + x_2 \leq 100 \\ & x_1, x_2 \geq 0 \end{aligned} \quad Q(x, \xi) = \begin{cases} \min & 7y_1 + 12y_2 \\ \text{s.t.} & y_1 \geq h_1 - (\alpha x_1 + 6x_2) \\ & y_2 \geq h_2 - (3x_1 + \beta x_2) \\ & y_1, y_2 \geq 0 \end{cases}$$



Dual formulation of $Q(x, \xi)$



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$$Q^{\text{primal}}(x, \xi) := \begin{cases} \min & q^\top y \\ \text{s.t.} & Wy = h - Tx \\ & y \geq 0 \end{cases}$$

- Dual problem of the LP:

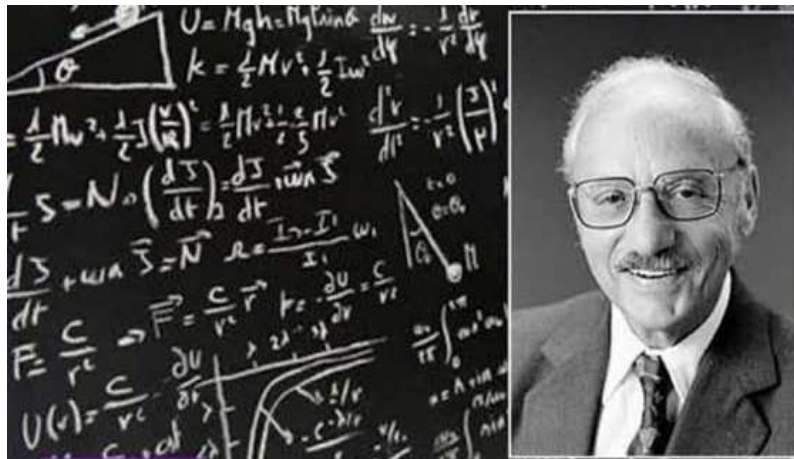
$$Q^{\text{dual}}(x, \xi) := \begin{cases} \max & \pi^\top (h - Tx) \\ \text{s.t.} & W^\top \pi \leq q \end{cases}$$

- From the **strong duality theory**:
 - Primal feasible and bounded $\rightarrow Q^{\text{dual}}(x, \xi) = Q^{\text{primal}}(x, \xi)$
 - Primal unbounded below $\rightarrow$ Dual infeasible
 - Primal infeasible $\rightarrow$ Dual unbounded above if feasible



The “Homework” Papers of G. B. Dantzig’s PhD Thesis:

- G. B. Dantzig. On the nonexistence of tests of “Students’ hypothesis having power functions independents of σ , *Annals of Mathematical Statistics* 11 (1940), 186-192.
- G. B. Dantzig and A. Wald. **On the fundamental Lemma of Neyman and Pearson**, *Annals of Mathematical Statistics*, 22 (1951), 87-93.





- Original (**sufficiency**) by JN/ESP, *Stat. Res. Memoires* 1936:

Suppose f_0, f_1 candidate functions to test, a binary test function $\phi(x)$ with a reject region R

$$\phi(x) = \begin{cases} 1 & \text{if } x \in R \\ 0 & \text{if } x \in R^c \end{cases}$$

$\phi(x)$ satisfies condition $\mathbf{P}_\alpha, \forall \alpha \in [0, 1]$

1) $\alpha = \mathbf{Pr}_{f_0}(x \in R) = \int f_0(x)\phi(x)dx$

2) $\exists \eta \geq 0$, such that

$$f_1(x) > \eta f_0(x), x \in R$$

$$f_1(x) < \eta f_0(x), x \in R^c$$



$\phi(x)$ is UMP of size α

for any other $\phi'(x)$ satisfying

$$\int f_0(x)\phi'(x)dx \leq \alpha$$

we have



$$\int f_1(x)\phi'(x)dx \leq \int f_1(x)\phi(x)dx$$



- Build the **necessity** by viewing it as an SP (SI LP) problem
 $\phi(x)$ is UMP of size α :

for any other $\phi'(x)$ satisfying

$$\int f_0(x)\phi'(x)dx \leq \alpha$$

we have

$$\int f_1(x)\phi'(x)dx \leq \int f_1(x)\phi(x)dx$$

$$\begin{aligned} \Rightarrow \max_{\phi \in \Phi} \mathbb{E}_{f_1}[\phi(x)] &= \int f_1(x)\phi(x)dx \\ \text{s.t. } \mathbb{E}_{f_0}[\phi(x)] &= \int f_0(x)\phi(x)dx \leq \alpha \end{aligned}$$



- The dual variable gives directly the likelihood ratio!

$$\begin{aligned} \max_{\phi \in \Phi} \quad & \mathbb{E}_{f_1}[\phi(x)] = \int f_1(x) \phi(x) dx \\ \text{s.t.} \quad & \mathbb{E}_{f_0}[\phi(x)] = \int f_0(x) \phi(x) dx \leq \alpha \quad \eta \geq 0 \end{aligned}$$

$$= \sup_{\phi} L(\phi, \eta^*) = \int [f_1(x) - \eta^* f_0(x)] \phi(x) dx + \eta^* \alpha$$

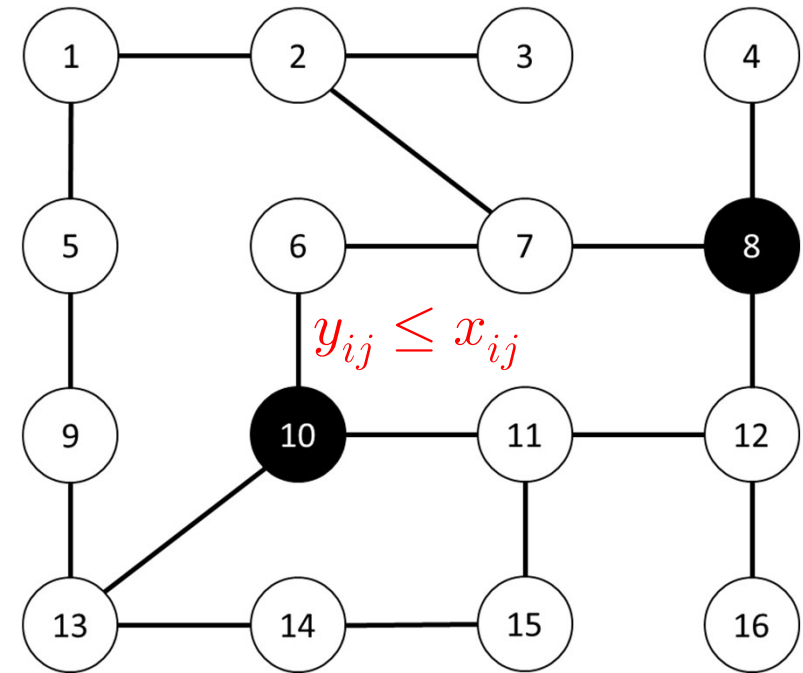
$$\Rightarrow \phi(x)^* = \begin{cases} 1 & \text{if } \frac{f_1(x)}{f_0(x)} > \eta^* \\ 0 & \text{if } \frac{f_1(x)}{f_0(x)} < \eta^* \end{cases}$$



$$\begin{aligned} \min \quad & x_1 + x_2 \\ \text{s.t.} \quad & \xi_1 x_1 + x_2 \geq 7 \\ & \xi_2 x_1 + x_2 \geq 4 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$$

- $\xi = (\xi_1, \xi_2) \in \Xi = \{(1, 1/3), (5/2, 2/3), (4, 1)\}$
- Each outcome has the same probability $p_s = 1/3$
- Does the problem make sense? How to reformulate the problem?

- White node set N_d with **uncertain demand** $\xi_n \geq 0$ for $n \in N_d$
 - Black node set N_p with supply capacity P_n
 - Bi-directional link set E , **capacity** x_{ij} **to design initially**, unit capacity investment cost c_{ij}
 - Unit link transport cost q_{ij}
 - Transport decisions y_{ij} will be done after realized demand
-



- **Questions:**
 - 1) formulate the 2nd stage recourse problem, minimizing total transport cost
 - 2) formulate the whole 2-stage SLP problem, minimizing total investment and expected transport cost



- 3) Does the problem have
 - fixed recourse?
 - complete fixed recourse?
 - relatively complete fixed recourse?
- 4) If no, how to reformulate the problem so that it has RCFR?



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